

TEST OF MATHEMATICS FOR UNIVERSITY ADMISSION

PAPER 2: MATHEMATICAL REASONING

Practice paper · Set 11

75 minutes

INSTRUCTIONS TO CANDIDATES

Please read these instructions carefully, but do not open the question paper until you are told that you may do so.

A separate answer sheet is provided for this paper. You require a soft pencil and an eraser. Please complete the answer sheet with your candidate number, centre number, date of birth and full name.

This paper contains **20 questions**. For each question, choose the **one** answer you consider correct and record your choice on the separate answer sheet. If you make a mistake, erase thoroughly and try again.

There are no penalties for incorrect responses, only marks for correct answers, so you should attempt all 20 questions. Each question is worth one mark.

You can use the question paper for rough working or notes, but no extra paper is allowed.

You must complete the answer sheet within the time limit.

Calculators and dictionaries are NOT permitted.

There is no formulae booklet for this test.

Please wait to be told you may begin before turning this page.

This is a practice paper generated from a question bank written to the TMUA content specification. It is not an official UAT-UK paper. Answers and worked solutions are in the accompanying solutions booklet.

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1 How many real solutions does the equation $x^4 - 5x^2 + 4 = 0$ have?

A 2

B 4

C 5

D 0

E 1

F 3

2 Solve $xy = 6$ and $x - y = 1$. What is the sum of all possible values of $x + y$?

A 10

B 5

C -5

D 7

E 1

F 0

3 Given that $2^x = 5$ and $2^y = 20$, find the value of 2^{2x-y} .

- A** $\frac{1}{4}$
- B** $\frac{5}{4}$
- C** $\frac{5}{4}$
- D** -10
- E** $\frac{25}{2}$
- F** 5

- 4 Let x be a real number. Which of the following statements is equivalent to ' $x \geq 2$ **if and only if** $x^3 \geq 8$ '?
- A** $x^3 \geq 8 \Rightarrow x \geq 2$
- B** $(x \geq 2 \Rightarrow x^3 \geq 8)$ and $(x^3 \geq 8 \Rightarrow x \geq 2)$
- C** $(x < 2 \Rightarrow x^3 \geq 8)$ or $(x^3 < 8 \Rightarrow x \geq 2)$
- D** $x \geq 2 \Rightarrow x^3 \geq 8$
- E** $x \geq 2$ or $x^3 \geq 8$

- 5 Consider the claim: **if n is a positive integer then $n^2 + n + 1$ is odd.** Which of the following is a valid proof?
- A** $n^2 + n = n(n + 1)$ is even, being the product of consecutive integers, so $n^2 + n + 1$ is even plus one, which is odd.
- B** $n^2 + n + 1$ cannot be factorised, so it is odd.
- C** For $n = 1, 2, 3$ the values are 3, 7, 13, all odd, so the claim holds.
- D** Suppose $n^2 + n + 1$ is even. Then n is even, which is a contradiction.
- E** If n is odd then n^2 is odd and n is odd, so $n^2 + n + 1$ is odd.

6 Claim: if $a > b$ then $\frac{1}{a} < \frac{1}{b}$.

I Suppose $a > b$.

II Dividing both sides by ab : $\frac{1}{b} > \frac{1}{a}$.

III So $\frac{1}{a} < \frac{1}{b}$.

Which of the following is correct?

- A The claim is false in general and the first error is on line II, since ab may be negative or zero.
- B The claim and proof are correct.
- C The first error is on line III.
- D The claim is true but the first error is on line I.
- E The claim is false and the first error is on line III.

- 7 Prove by cases that $n^3 - n$ is divisible by 3 for every integer n . Which set of cases is both sufficient and necessary to consider?
- A $n = 3k + 1, n = 3k + 2$
 - B n even, n odd
 - C n positive, n zero, n negative
 - D $n = 3k, n = 3k + 1, n = 3k + 2$
 - E n prime, n not prime

8 A student solves $2\cos^2 x = \cos x$ for $0 \leq x < 2\pi$.

I Divide both sides by $\cos x$: $2\cos x = 1$.

II So $\cos x = \frac{1}{2}$.

III So $x = \frac{\pi}{3}$ or $x = \frac{5\pi}{3}$.

How many solutions has the student missed, and where is the first error?

- A** Two solutions missed; the first error is on line I.
- B** No solutions missed; the working is correct.
- C** One solution missed; the first error is on line III.
- D** Two solutions missed; the first error is on line III.
- E** One solution missed; the first error is on line I.

9 Which of the following statements about the graph of $y = |2x - 4|$ is/are true?

- I** It meets the x -axis at exactly one point.
- II** It is symmetric about the line $x = 2$.
- III** It lies entirely on or above the line $y = 0$.

- A** none of them
- B** I only
- C** II only
- D** III only
- E** I and II only
- F** I and III only
- G** II and III only
- H** I, II and III

10 Consider the equation $\sin x = k$ where k is a real constant. Which of the following statements is/are true?

- I** For every k with $-1 < k < 1$ the equation has exactly two solutions in $0 \leq x < 2\pi$.
- II** For $k = 1$ the equation has exactly one solution in $0 \leq x < 2\pi$.
- III** For every real k the equation has at least one solution.

- A** none of them
- B** I only
- C** II only
- D** III only
- E** I and II only
- F** I and III only
- G** II and III only
- H** I, II and III

- 11 Which of the following is a counterexample to the claim '**for all positive x and y ,**
 $\log_{10}(x+y) \leq \log_{10} x + \log_{10} y$ '?
- A** $x = 10, y = 100$
- B** $x = y = 2$
- C** $x = 5, y = 5$
- D** $x = 4, y = 4$
- E** $x = y = 1$

12 The cubic $y = x^3 + ax + b$ has three distinct real roots. Which of the following statements must be true?

I $a < 0$

II $b \neq 0$

III The curve has two turning points.

A none of them

B I only

C II only

D III only

E I and II only

F I and III only

G II and III only

H I, II and III

- 13** A rectangle has integer side lengths and its area (in cm^2) is numerically equal to its perimeter (in cm). How many such rectangles are there (counting $a \times b$ and $b \times a$ once)?
- A** 4
 - B** 3
 - C** 0
 - D** 1
 - E** 2
 - F** infinitely many

- 14** Which of the following statements about the graph of $y = \cos x$ is/are true?
- I** It is symmetric about the y -axis.
 - II** It is a translation of the graph of $y = \sin x$ by $\frac{\pi}{2}$ in the negative x -direction.
 - III** It has period π .
- A** none of them
- B** I only
- C** II only
- D** III only
- E** I and II only
- F** I and III only
- G** II and III only
- H** I, II and III

- 15** A sequence of 10 integers is such that the sum of any three consecutive terms is positive and the sum of any five consecutive terms is negative. Which of the following statements must be true?

- I** The sum of all 10 terms is negative.
- II** The sequence contains at least one negative term.
- III** The sequence contains at least one positive term.

- A** none of them
- B** I only
- C** II only
- D** III only
- E** I and II only
- F** I and III only
- G** II and III only
- H** I, II and III

16 Which of the following statements is/are true about real numbers?

I For all x and all y , if $x < y$ then $x^2 < y^2$.

II For all x there exists y such that $x < y$ and $x^2 < y^2$.

III There exist x and y with $x < y$ and $x^2 > y^2$.

A none of them

B I only

C II only

D III only

E I and II only

F I and III only

G II and III only

H I, II and III

17 The quadratic equation $ax^2 + bx + c = 0$ (with $a \neq 0$) has real coefficients. Which of the following statements is/are always true?

- I** If $ac < 0$, the equation has two distinct real roots.
- II** If $b = 0$, the equation has no real roots.
- III** If the equation has a repeated root, then $b^2 = 4ac$.

- A** none of them
- B** I only
- C** II only
- D** III only
- E** I and II only
- F** I and III only
- G** II and III only
- H** I, II and III

- 18** Consider the claim: **if x and y are real numbers with $x + y > 10$, then $x > 5$ or $y > 5$.** Which of the following proves it?
- A** Suppose $x \leq 5$ and $y \leq 5$. Then $x + y \leq 10$, contradicting $x + y > 10$.
- B** Suppose $x > 5$. Then $y > 10 - x$, so $y > 5$.
- C** If $x + y > 10$ then the average of x and y is more than 5, so both exceed 5.
- D** Take $x = 6$, $y = 6$: then $x + y = 12 > 10$ and $x > 5$.
- E** If $x > 5$ then the conclusion holds; if $y > 5$ then the conclusion holds.

19 Which of the following statements is/are logically equivalent to '**if it is raining, then the ground is wet**'?

- I** It is not raining, or the ground is wet.
- II** If the ground is not wet, then it is not raining.
- III** If the ground is wet, then it is raining.

- A** none of them
- B** I only
- C** II only
- D** III only
- E** I and II only
- F** I and III only
- G** II and III only
- H** I, II and III

- 20** Let f be a polynomial. P : ' $f'(x) > 0$ for all real x '. Q : ' $f(x) = 0$ has exactly one real root'. Which one of the following is true?
- A** P is necessary and sufficient for Q .
 - B** P is necessary but not sufficient for Q .
 - C** P is sufficient but not necessary for Q .
 - D** P is neither necessary nor sufficient for Q .

END OF TEST